

Executive Summary

Open Neighbourhoods Precede Concept Spread: Structural Antecedents and Cross-Disciplinary Diffusion of Emerging Scientific Concepts

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Headline

When a scientific concept enters a new subfield, the share of its initial co-occurrence partners that are native to that host predicts whether the concept will be taken up by the host’s own newcomer authors. The effect replicates on a held-out concept fold (IRR/SD 1.19 [1.06, 1.33], $p = 0.004$) and an independent MeSH biomedical population (IRR/SD 1.23 [1.12, 1.36], Holm $p < 0.001$), pools to IRR/SD 1.26 [1.17, 1.36], operates through both extensive and intensive margins, and is host-specific rather than reflecting general concept accessibility.

Goal

The study investigates two questions using an OpenAlex-based scholarly dataset of 426 semantically grounded concepts with 462,812 works:

RQ1 (Emergence) Which temporal and structural patterns in an evolving, semantically grounded scientific knowledge graph characterise the emergence of scientific concepts?

RQ2 (Diffusion) How do emerging scientific concepts diffuse across disciplinary communities over time, and which temporal network patterns distinguish locally concentrated concepts from broadly integrated ones?

The target venue is *Applied Network Science*.

What Was Tried

Five iterations over 25 artifacts:

1. **Data foundation (Iteration 1).** Built a concept pool of 426 concepts (366 main, 60 reference), 462,812 works, and 488,078 verified concept-work links from OpenAlex with a held-out MeSH confirmation population of 191 biomedical concepts. Pre-registered all decision rules and hashed specifications before data inspection.
2. **Gates, viability, and emergence event study (Iteration 2).** Tested the citation-layer gate (Gate A) at edge level; built 25 yearly co-word network snapshots ($\sim 27,000$ nodes, $\sim 84,000$ edges); ran the RQ1 precursor event study on the screen fold and MeSH population. Originally planned source-sink viability estimation; blocked by Gate A failure.

3. **Deepen closure, test grafting, describe diffusion (Iteration 3).** Deepened the closure finding with turnover-proof openness measures. Tested host-entry grafting on 1,740 screen-fold entries using PPML regression. Built a $k=2$ diffusion typology and tested expansion-before-diffusion ordering.
4. **Held-out confirmation and replication (Iteration 4).** Opened four sealed held-out folds. Confirmed the co-primary grafting specification. Replicated on MeSH. Tested single-paper artefact, vocabulary decomposition, classifier circularity, closure-anchoring link, and adopter-level mechanism.
5. **Post-confirmation decomposition, audit, positioning (Iteration 5).** Ran three K-tests: margin decomposition (both extensive and intensive confirmed), host-specificity (confirmed), and field boundary (no variation). Full numerical audit (319 curated numbers, 99.4% accuracy). Positioning dossier grading novelty against 25 candidates.

Key Results

Table 1: Summary of confirmed and dead findings.

Finding	Evidence	Status
Host-vocabulary grafting (RQ2)	Co-primary IRR/SD: screen 1.30, held-out 1.19, MeSH 1.23; IVW pooled 1.26 [1.17, 1.36]	CONFIRMED
Margin decomposition (K1)	Extensive IVW +6.43 pp/SD; intensive IVW IRR/SD 1.183	BOTH
Host-specificity (K2)	IVW retention 0.969; A_{lift} IVW IRR/SD 1.34 [1.23, 1.46]	HOST-SPECIFIC
Field boundary (K3)	Wald $p = 0.638$	NO VARIATION
Persistent-neighbour closure (RQ1)	Held-out $S = -1.073$, Holm $p = 0.015$	CONFIRMED
Diffusion typology ($k=2$)	Localised (136) vs broad-from-start (66); replicates on held-out	CONFIRMED
Expansion before diffusion	25 of 26 pooled dual-onset concepts	CONFIRMED
Adopter enrichment	OR 3.09 [2.32, 4.28]; gradient FOREIGN > ADJACENT > NATIVE	CONFIRMED
Primary closure (R1)	Holm $p = 0.147$ on held-out	R1_DEAD
Source-sink viability (C0)	Gate A failure: within-host traced share 0.17	BLOCKED
Breadth prediction	All Holm $p = 0.405$	NOT SUPPORTED
Closure-anchoring link	Partial $\rho = 0.049$, $p = 0.564$	NULL

What Failed

- **Gate A (citation-layer feasibility).** Only 17.3% of host edge-years meet the within-host tracing threshold (vs 64.2% under lenient any-parent tracing). The viability decomposition central to the original hypothesis could not be executed.
- **Primary closure (R1_DEAD).** The general closure deficit does not survive held-out confirmation (Holm $p = 0.147$). Only the persistent-neighbour variant survives.

- **Breadth prediction.** Openness does not predict where concepts travel beyond growth baselines.
- **Community roles.** Leiden community roles (BRIDGE, STAYER, MIGRANT) do not predict host-subfield entry (BRIDGE OR 0.64 [0.38, 1.08]).
- **Co-transfer.** Arriving with origin companions is null across all folds and populations.
- **Concept pool field coverage.** 363 of 366 eligible candidates from the arXiv arm; social sciences and humanities are absent.

Open Questions

1. **Generalisability.** The population is dominated by physics/CS arXiv concepts. Extension to social sciences and humanities is untested.
2. **Primary specification power.** The fully-saturated primary specification is underpowered on the held-out fold and MeSH. Only the co-primary specification supports the grafting claim.
3. **Establishment threshold null.** The binary establishment threshold (EST_{bin}) is null on the held-out fold (+2.54 pp, $p = 0.16$).
4. **Single-paper dominance.** 87% of screen co-primary events are single-paper entries; the held-out interaction ($p = 0.004$) suggests the effect is stronger for single-paper entries.
5. **Adopter-vs-entry paradox.** The adopter-level vocabulary gradient (FOREIGN > NATIVE) runs opposite to the entry-level finding (host-native partners predict uptake). Whether these reflect different mechanisms at different scales is unresolved.
6. **Audit residuals.** 292 numbers in artifact outputs were not transcribed into the report. Seeded-perturbation recall is 0.60.